

Experiment-8

Customer Churn Prediction using Decision Tree Classification

```
In [1]: # Step 1: Load Dataset

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix,
    classification_report
)
from sklearn.tree import plot_tree

df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")

print("First 5 Records:")
print(df.head())

# Step 2: Explore Dataset

print("\nDataset Shape:")
print(df.shape)

print("\nDataset Information:")
print(df.info())

print("\nStatistical Summary:")
print(df.describe())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Values:")
print(df.duplicated().sum())

# Step 3: Data Preprocessing

df = df.drop_duplicates()

df["TotalCharges"] = pd.to_numeric(
    df["TotalCharges"],
    errors="coerce"
)

df["TotalCharges"] = df["TotalCharges"].fillna(
```

```
        df["TotalCharges"].median()
    )

    print("\nMissing Values After Preprocessing:")
    print(df.isnull().sum())

    # Step 4: Remove Unnecessary Columns

    df = df.drop(
        columns=["customerID"]
    )

    print("\nColumns After Removing Customer ID:")
    print(df.columns)

    # Step 5: Encode Categorical Variables

    categorical_columns = df.select_dtypes(
        include=["object"]
    ).columns

    encoder = LabelEncoder()

    for column in categorical_columns:
        df[column] = encoder.fit_transform(
            df[column]
        )

    print("\nEncoded Dataset:")
    print(df.head())

    # Step 6: Select Features and Target

    X = df.drop(
        columns=["Churn"]
    )

    y = df["Churn"]

    print("\nFeatures:")
    print(X.columns)

    print("\nTarget:")
    print(y.name)

    # Step 7: Split Dataset into Training and Testing Data

    X_train, X_test, y_train, y_test = train_test_split(
        X,
        y,
        test_size=0.20,
        random_state=42,
        stratify=y
    )

    print("\nTraining Data Shape:")
```

```
print(X_train.shape)

print("\nTesting Data Shape:")
print(X_test.shape)

# Step 8: Train Decision Tree Classifier

model = DecisionTreeClassifier(
    criterion="gini",
    max_depth=5,
    random_state=42
)

model.fit(
    X_train,
    y_train
)

print("\nDecision Tree Model Trained Successfully.")

# Step 9: Predict Customer Churn

y_pred = model.predict(
    X_test
)

print("\nPredicted Values:")
print(y_pred[:20])

# Step 10: Evaluate Model

accuracy = accuracy_score(
    y_test,
    y_pred
)

precision = precision_score(
    y_test,
    y_pred
)

recall = recall_score(
    y_test,
    y_pred
)

f1 = f1_score(
    y_test,
    y_pred
)

print("\nModel Evaluation:")

print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1 Score :", round(f1, 4))
```

```
# Step 11: Classification Report

print("\nClassification Report:")

print(
    classification_report(
        y_test,
        y_pred
    )
)

# Step 12: Confusion Matrix

cm = confusion_matrix(
    y_test,
    y_pred
)

print("\nConfusion Matrix:")
print(cm)

plt.figure(figsize=(7, 5))

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["No Churn", "Churn"],
    yticklabels=["No Churn", "Churn"]
)

plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.show()

# Step 13: Visualize Decision Tree

plt.figure(figsize=(20, 10))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=["No Churn", "Churn"],
    filled=True,
    rounded=True,
    fontsize=8
)

plt.title("Decision Tree for Customer Churn Prediction")

plt.show()
```

```
# Step 14: Find Feature Importance

feature_importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": model.feature_importances_
})

feature_importance = feature_importance.sort_values(
    by="Importance",
    ascending=False
)

print("\nFeature Importance:")
print(feature_importance)

# Step 15: Visualize Feature Importance

plt.figure(figsize=(10, 6))

sns.barplot(
    data=feature_importance,
    x="Importance",
    y="Feature"
)

plt.title("Feature Importance in Customer Churn Prediction")
plt.xlabel("Importance")
plt.ylabel("Feature")

plt.show()

# Step 16: Analyze Churn Distribution

churn_count = df["Churn"].value_counts()

print("\nCustomer Churn Distribution:")
print(churn_count)

plt.figure(figsize=(6, 5))

sns.countplot(
    x=df["Churn"]
)

plt.title("Customer Churn Distribution")
plt.xlabel("Churn")
plt.ylabel("Number of Customers")

plt.show()

# Step 17: Display Customers Predicted to Churn

churn_predictions = X_test.copy()

churn_predictions["Actual Churn"] = y_test.values
churn_predictions["Predicted Churn"] = y_pred
```

```
print("\nCustomers Predicted to Churn:")

print(
    churn_predictions[
        churn_predictions["Predicted Churn"] == 1
    ].head(20)
)

# Step 18: Final Result

print("\nFinal Result:")

print(
    "Decision Tree Customer Churn Prediction completed successfully."
)

print(
    f"Model Accuracy: {accuracy * 100:.2f}%"
)

print(
    f"Precision: {precision * 100:.2f}%"
)

print(
    f"Recall: {recall * 100:.2f}%"
)

print(
    f"F1 Score: {f1 * 100:.2f}%"
)
```

First 5 Records:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	\
0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	

	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	\
0	No phone service	DSL	No	...	No	
1	No	DSL	Yes	...	Yes	
2	No	DSL	Yes	...	No	
3	No phone service	DSL	Yes	...	Yes	
4	No	Fiber optic	No	...	No	

	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	\
0	No	No	No	Month-to-month	Yes	
1	No	No	No	One year	No	
2	No	No	No	Month-to-month	Yes	
3	Yes	No	No	One year	No	
4	No	No	No	Month-to-month	Yes	

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

[5 rows x 21 columns]

Dataset Shape:
(7043, 21)

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 7043 entries, 0 to 7042

Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
0	customerID	7043 non-null	object
1	gender	7043 non-null	object
2	SeniorCitizen	7043 non-null	int64
3	Partner	7043 non-null	object
4	Dependents	7043 non-null	object
5	tenure	7043 non-null	int64
6	PhoneService	7043 non-null	object
7	MultipleLines	7043 non-null	object
8	InternetService	7043 non-null	object
9	OnlineSecurity	7043 non-null	object
10	OnlineBackup	7043 non-null	object
11	DeviceProtection	7043 non-null	object
12	TechSupport	7043 non-null	object
13	StreamingTV	7043 non-null	object
14	StreamingMovies	7043 non-null	object
15	Contract	7043 non-null	object
16	PaperlessBilling	7043 non-null	object
17	PaymentMethod	7043 non-null	object
18	MonthlyCharges	7043 non-null	float64
19	TotalCharges	7043 non-null	object

20 Churn 7043 non-null object
 dtypes: float64(1), int64(2), object(18)
 memory usage: 1.1+ MB
 None

Statistical Summary:

	SeniorCitizen	tenure	MonthlyCharges
count	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692
std	0.368612	24.559481	30.090047
min	0.000000	0.000000	18.250000
25%	0.000000	9.000000	35.500000
50%	0.000000	29.000000	70.350000
75%	0.000000	55.000000	89.850000
max	1.000000	72.000000	118.750000

Missing Values:

customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0

dtype: int64

Duplicate Values:

0

Missing Values After Preprocessing:

customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0

```
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

Columns After Removing Customer ID:

```
Index(['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure',
      'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity',
      'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV',
      'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod',
      'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='object')
```

Encoded Dataset:

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	\
0	0	0	1	0	1	0	
1	1	0	0	0	34	1	
2	1	0	0	0	2	1	
3	1	0	0	0	45	0	
4	0	0	0	0	2	1	

	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	\
0	1	0	0	2	
1	0	0	2	0	
2	0	0	2	2	
3	1	0	2	0	
4	0	1	0	0	

	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	\
0	0	0	0	0	0	
1	2	0	0	0	1	
2	0	0	0	0	0	
3	2	2	0	0	1	
4	0	0	0	0	0	

	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	1	2	29.85	29.85	0
1	0	3	56.95	1889.50	0
2	1	3	53.85	108.15	1
3	0	0	42.30	1840.75	0
4	1	2	70.70	151.65	1

Features:

```
Index(['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure',
      'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity',
      'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV',
      'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod',
      'MonthlyCharges', 'TotalCharges'],
      dtype='object')
```

Target:

Churn

Training Data Shape:

(5634, 19)

Testing Data Shape:

(1409, 19)

Decision Tree Model Trained Successfully.

Predicted Values:

[0 1 0 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0]

Model Evaluation:

Accuracy : 0.785

Precision: 0.6035

Recall : 0.5535

F1 Score : 0.5774

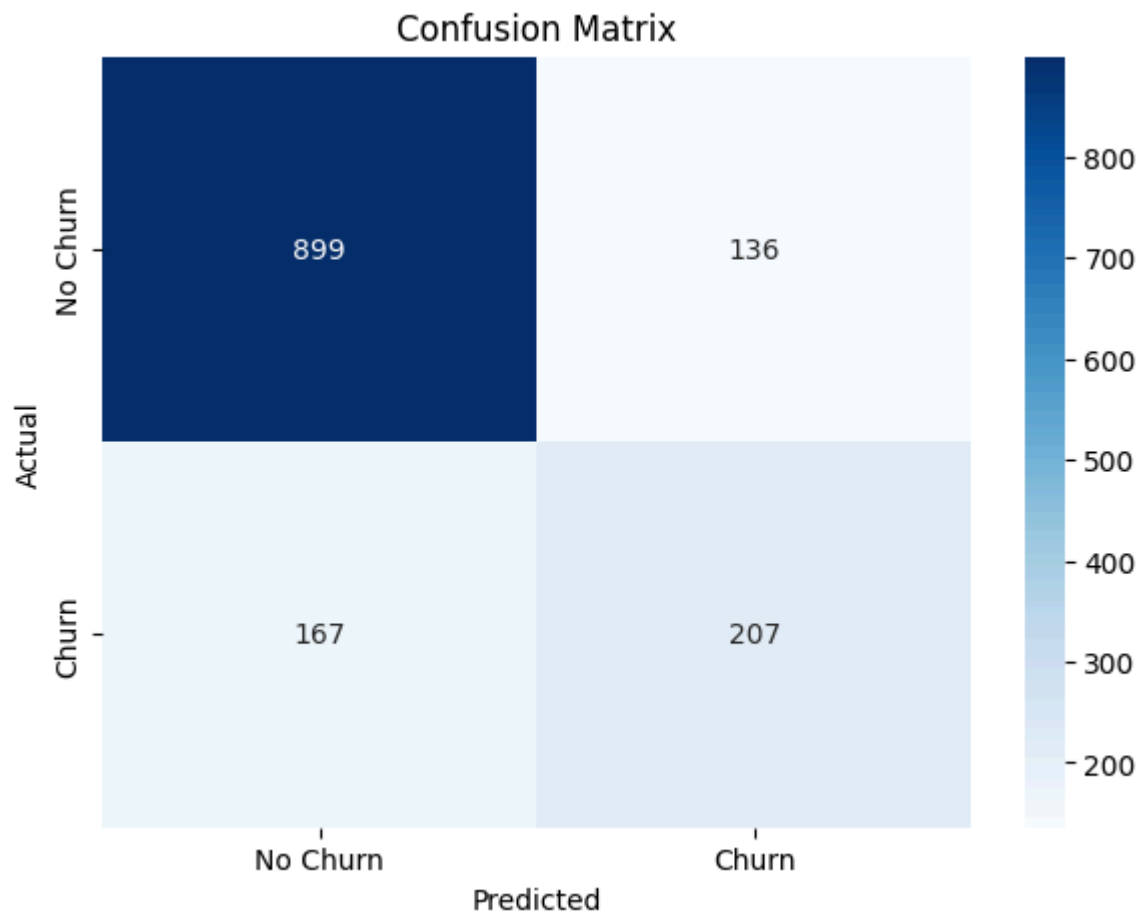
Classification Report:

	precision	recall	f1-score	support
0	0.84	0.87	0.86	1035
1	0.60	0.55	0.58	374
accuracy			0.78	1409
macro avg	0.72	0.71	0.72	1409
weighted avg	0.78	0.78	0.78	1409

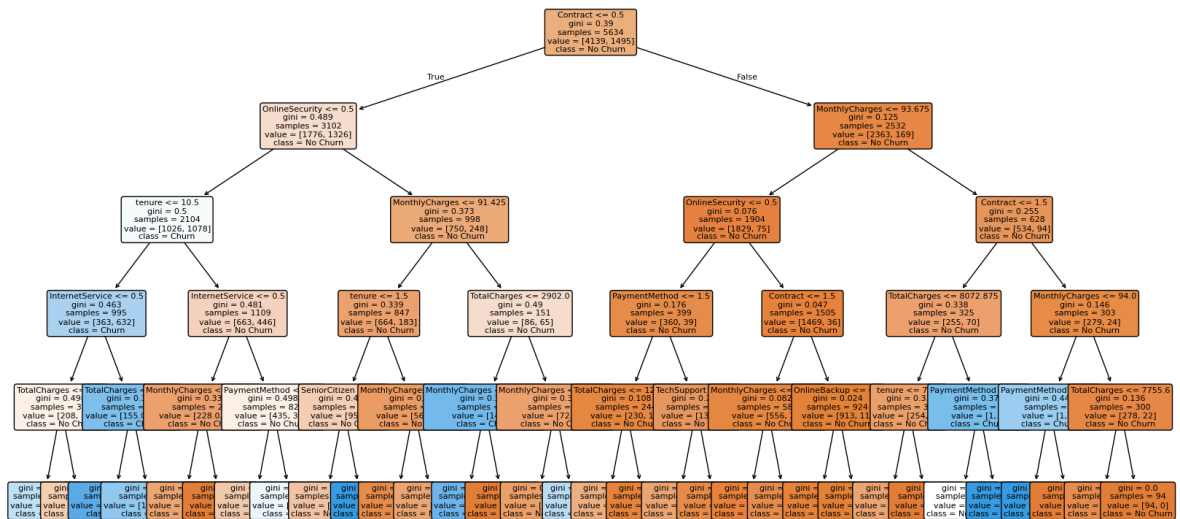
Confusion Matrix:

[[899 136]

[167 207]]



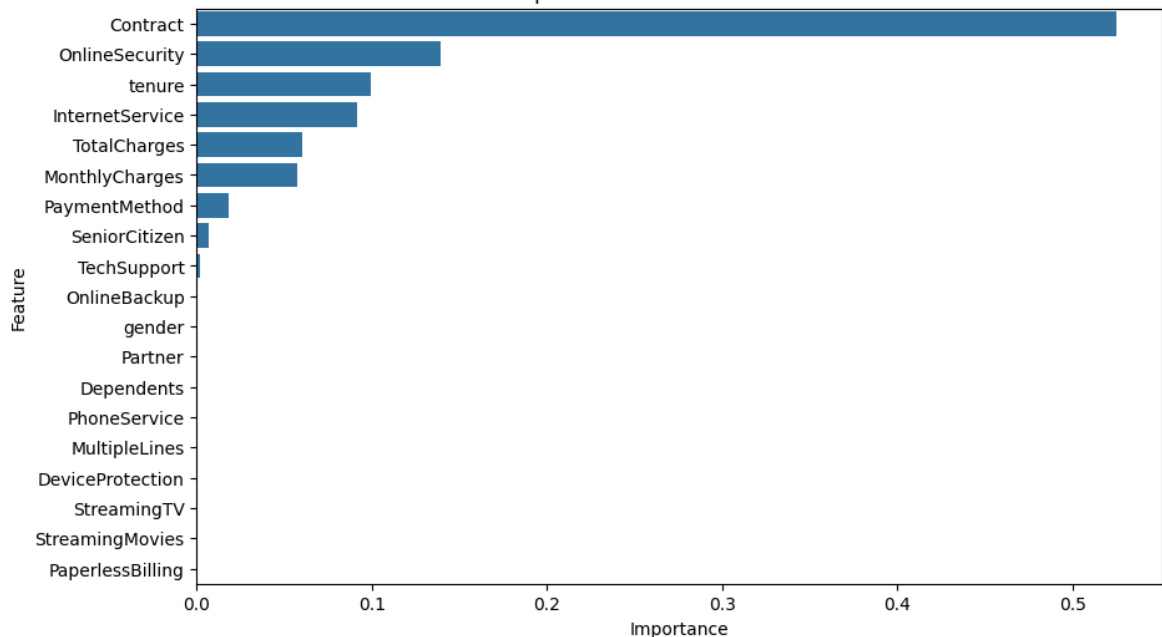
Decision Tree for Customer Churn Prediction



Feature Importance:

	Feature	Importance
14	Contract	0.525341
8	OnlineSecurity	0.138973
4	tenure	0.099174
7	InternetService	0.091187
18	TotalCharges	0.060525
17	MonthlyCharges	0.057434
16	PaymentMethod	0.018331
1	SeniorCitizen	0.006759
11	TechSupport	0.002142
9	OnlineBackup	0.000135
0	gender	0.000000
2	Partner	0.000000
3	Dependents	0.000000
5	PhoneService	0.000000
6	MultipleLines	0.000000
10	DeviceProtection	0.000000
12	StreamingTV	0.000000
13	StreamingMovies	0.000000
15	PaperlessBilling	0.000000

Feature Importance in Customer Churn Prediction



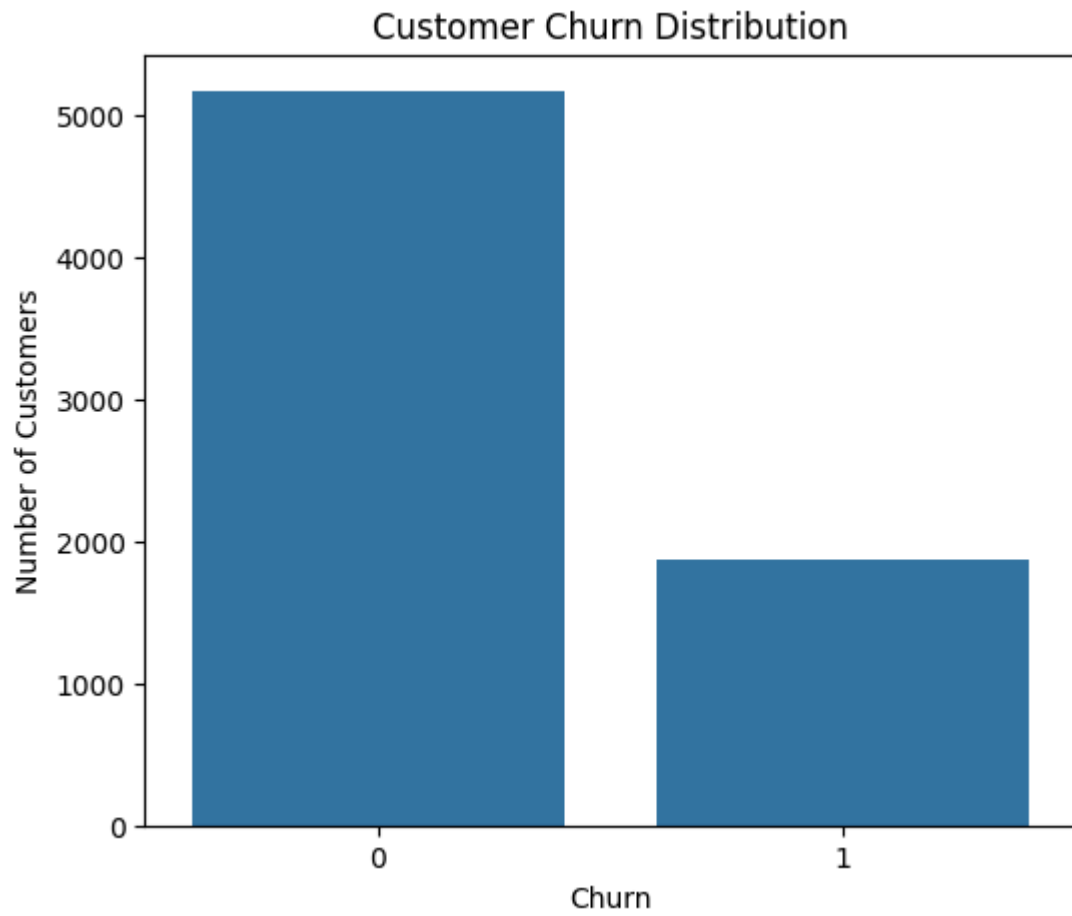
Customer Churn Distribution:

Churn

0 5174

1 1869

Name: count, dtype: int64



Customers Predicted to Churn:

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	\
2280	0	1	0	0	8	1	
4460	1	0	1	0	18	1	
2136	1	0	0	1	15	1	
6529	1	1	0	0	2	1	
5389	0	0	1	0	51	1	
761	0	0	0	0	22	1	
5547	0	0	0	0	3	1	
1838	1	1	0	0	54	1	
6979	1	0	0	1	1	0	
346	0	0	0	0	2	1	
6633	0	0	0	0	1	1	
2519	1	0	1	0	2	1	
3510	0	0	0	0	8	1	
2246	0	0	1	1	1	1	
6304	0	0	0	0	9	1	
5300	0	0	1	1	1	1	
290	1	1	0	0	24	1	
642	1	0	0	0	1	1	
3856	1	0	1	0	61	1	
6314	0	0	0	0	6	1	

	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	...	\
2280	2	1	0	0	...	
4460	0	1	0	0	...	
2136	2	1	0	2	...	
6529	2	1	0	0	...	
5389	0	1	0	0	...	
761	0	1	0	0	...	
5547	0	1	0	0	...	
1838	2	1	0	2	...	
6979	1	0	0	0	...	
346	0	1	0	0	...	
6633	2	1	0	0	...	
2519	2	1	0	0	...	
3510	2	1	0	0	...	
2246	2	1	2	0	...	
6304	0	1	0	0	...	
5300	0	1	0	0	...	
290	2	1	0	0	...	
642	0	1	0	0	...	
3856	2	1	0	2	...	
6314	2	1	0	0	...	

	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	\
2280	2	2	2	0	1	
4460	2	0	0	0	0	
2136	0	2	0	0	1	
6529	0	0	0	0	1	
5389	0	0	0	0	0	
761	0	2	2	0	0	
5547	0	2	2	0	1	
1838	0	0	2	0	1	
6979	0	0	0	0	0	
346	0	2	2	0	1	
6633	0	0	0	0	1	
2519	0	2	0	0	1	
3510	0	0	0	0	1	
2246	0	2	2	0	1	

6304	0	2	0	0	1
5300	0	2	0	0	1
290	0	0	0	0	1
642	0	2	2	0	1
3856	2	2	2	0	1
6314	0	0	0	0	1

	PaymentMethod	MonthlyCharges	TotalCharges	Actual Churn	\
2280	1	100.15	908.55	0	
4460	2	78.20	1468.75	0	
2136	2	91.00	1430.05	0	
6529	3	74.85	156.40	1	
5389	3	69.15	3649.60	0	
761	2	89.25	1907.85	1	
5547	2	89.45	240.45	1	
1838	2	90.05	4931.80	0	
6979	3	24.20	24.20	0	
346	2	90.40	168.20	1	
6633	2	74.50	74.50	1	
2519	2	91.45	171.45	0	
3510	2	75.25	576.70	0	
2246	2	102.45	102.45	1	
6304	0	77.65	714.15	1	
5300	3	80.05	80.05	1	
290	2	76.10	1712.70	1	
642	2	89.55	89.55	1	
3856	2	114.10	7132.15	0	
6314	3	73.85	401.30	0	

	Predicted Churn
2280	1
4460	1
2136	1
6529	1
5389	1
761	1
5547	1
1838	1
6979	1
346	1
6633	1
2519	1
3510	1
2246	1
6304	1
5300	1
290	1
642	1
3856	1
6314	1

[20 rows x 21 columns]

Final Result:

Decision Tree Customer Churn Prediction completed successfully.

Model Accuracy: 78.50%

Precision: 60.35%

Recall: 55.35%

F1 Score: 57.74%